# Knox Design Analysis Notebook

Spring 2026

EC552 - Computational Synthetic Biology for Engineers

Team Members Names:

# Notes From James

Depending on number of rules, the rule evaluation algorithm can take 10-30 minutes to run. Rule Evaluations are saved in Neo4j, so you do not need to rerun the rule evaluation algorithm, call getRuleEvaluation with the name of the evaluation (only 2-5 seconds to retrieve).

Run rule evaluations with:

- ruleEvaluateByGroup
- ruleEvaluateByDesigns

Get rule evaluations with

- getRuleEvaluation

Delete rule evaluations with

- deleteRuleEvaluation

This notebook is to serve as a starting point for doing your analysis.

Feel free to change this notebook in any way you see fit.

```
import requests
import csv
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import json

from sklearn import tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, mean_squared_error, classification_
```

```
url = 'http://localhost:8080'

def ruleEvaluateByGroup(evalName, groupID, ruleGroupID, labelingMethod="sign"):
    """
    API Request to Knox to run Rule Evaluation Algorithm.

    Returns:
    metrics (list of pandas DataFrames): purity_metrics, designToRule
    """
    response = requests.post(url + '/rule/evaluate?' +
                             "evaluationName=" + evalName + '&' +
                             "designGroupID=" + groupID + '&' +
                             "rulesGroupID=" + ruleGroupID + '&' +
                             "labelingMethod=" + labelingMethod
    )

    return processRuleEval(response)


def ruleEvaluateByDesigns(evalName, designIDs, ruleGroupID, designScores, label
    designSpaceIDs = listToStringList(designIDs)
    designScoresStr = listToStringList(designScores) # designScores should be a

    # Submit Request
    response = requests.post(url + '/rule/evaluate?' +
                             "evaluationName=" + evalName + '&' +
                             "designSpaceIDs=" + designSpaceIDs + '&' +
                             "rulesGroupID=" + ruleGroupID + '&' +
                             "designScores=" + designScoresStr + '&' +
                             "labelingMethod=" + labelingMethod
    )

    return processRuleEval(response)


def getRuleEvaluation(evalName):
    response = requests.get(url + '/rule/getEvaluation?' + "evaluationName=" +

    return processRuleEval(response)


def deleteRuleEvaluation(evalName):
    response = requests.delete(url + '/rule?' + "evaluationName=" + evalName)

    if not response.text:
        return f'"{evalName}" Sucessfully Deleted'
    else:
        return response.text


def processRuleEval(response):
```

```
    # Change to Pandas DataFrame
    json_data = json.loads(response.text)

    purity_metrics_df = pd.DataFrame(json_data["evaluationResults"]).T

    designToRule_df = pd.DataFrame(json_data["designToRule"], index=json_data["
    cols = designToRule_df.columns.to_list()
    cols.remove("labels")
    cols.remove("scores")
    cols.remove("designIDs")
    cols = ["labels", "scores"] + cols
    designToRule_df = designToRule_df[cols]

    return purity_metrics_df.sort_values("impact"), designToRule_df.sort_values


def listToStringList(list_input):
    return ",".join(list_input)
```

```
def exampleTree(X, y, **kwargs):
    """Binary / multi-class classification tree."""
    dt_clf = tree.DecisionTreeClassifier(
        splitter=kwargs.get('splitter', 'best'),
        max_depth=kwargs.get('max_depth', None),
        min_samples_split=kwargs.get('min_samples_split', 400),
        min_samples_leaf=kwargs.get('min_samples_leaf', 200),
        max_features=kwargs.get('max_features', None),
        max_leaf_nodes=kwargs.get('max_leaf_nodes', None)
    )
    dt_clf = dt_clf.fit(X, y)
    return dt_clf


def regressionTree(X, y, **kwargs):
    """Regression tree predicting continuous scores."""
    dt_reg = tree.DecisionTreeRegressor(
        splitter=kwargs.get('splitter', 'best'),
        max_depth=kwargs.get('max_depth', None),
        min_samples_split=kwargs.get('min_samples_split', 400),
        min_samples_leaf=kwargs.get('min_samples_leaf', 200),
        max_features=kwargs.get('max_features', None),
        max_leaf_nodes=kwargs.get('max_leaf_nodes', None)
    )
    dt_reg = dt_reg.fit(X, y)
    return dt_reg


def plotTree(model, feature_names, class_names=None, title="Decision Tree", sav
```

```
    """Plot and optionally save a fitted sklearn decision tree."""
    fig, ax = plt.subplots(figsize=figsize)
    tree.plot_tree(
        model,
        feature_names=feature_names,
        class_names=class_names,
        filled=True,
        rounded=True,
        fontsize=7,
        ax=ax
    )
    ax.set_title(title, fontsize=14)
    plt.tight_layout()
    if save_path:
        plt.savefig(save_path, dpi=150, bbox_inches='tight')
        print(f"Saved: {save_path}")
    plt.show()
```

```
# Use this for (gnn_predicted_scores.csv) designs already have the attached scc

evalName      = 'hw1_eval'        # any unique name — gets saved in Neo4j
groupID       = 'hw1_designs'     # Group ID you used when importing the 3 CSVs
ruleGroupID   = 'hw1_rules'       # Group ID you used in GOLDBAR Generator
labelingMethod = 'median'

purity_metrics_df, designToRule_df = ruleEvaluateByGroup(evalName, groupID, rul
```

```
#purity_metrics_df, designToRule_df = getRuleEvaluation('hw1_eval')
```

```
purity_metrics_df.to_csv("purity_metrics.csv")
```

```
designToRule_df.to_csv("designToRule.csv")
```

```
feature_names = designToRule_df.columns.to_list()[2:]   # rule column names
X             = designToRule_df.iloc[:, 2:].to_numpy(dtype=int)   # 0=kept, 1=e
y_labels      = designToRule_df["labels"].to_numpy(dtype=int)     # 0=poor, 1=g
y_scores      = designToRule_df["scores"].to_numpy(dtype=float)   # raw Cello s

print(f"Features (X):  {X.shape}")
print(f"Labels  (y):   {y_labels.shape}  |  unique: {np.unique(y_labels)}")
print(f"Scores  (y):   {y_scores.shape}  |  range [{y_scores.min():.3f}, {y_scc
print(f"Class balance — poor (0): {(y_labels==0).sum()}  |  good (1): {(y_label
```

```
Features (X):  (7992, 5)
Labels  (y):   (7992,)  |  unique: [0 1]
```

```
Scores  (y):   (7992,)  |  range [-0.955, 3.462]
Class balance — poor (0): 3996  |  good (1): 3996
```

```
RANDOM_STATE = 42
TEST_SIZE    = 0.30

# ── For binary classification and regression ──
X_train, X_test, y_train_lbl, y_test_lbl, y_train_scr, y_test_scr = train_test_
    X, y_labels, y_scores, test_size=TEST_SIZE, random_state=RANDOM_STATE
)

# ── For multi-class: bin scores into quartiles ──
q1, q2, q3 = np.quantile(y_scores, [0.25, 0.50, 0.75])
print(f"Score quartiles  Q1={q1:.3f}  Q2(median)={q2:.3f}  Q3={q3:.3f}")

def bin_to_quartile(s):
    if   s <= q1: return 0  # awful
    elif s <= q2: return 1  # poor
    elif s <= q3: return 2  # good
    else:         return 3  # great

y_multi = np.array([bin_to_quartile(s) for s in y_scores])
class_names_multi = ["awful", "poor", "good", "great"]

_, _, y_train_multi, y_test_multi = train_test_split(
    X, y_multi, test_size=TEST_SIZE, random_state=RANDOM_STATE
)

print(f"Train size: {X_train.shape[0]}  |  Test size: {X_test.shape[0]}")
```

```
Score quartiles  Q1=-0.628  Q2(median)=-0.077  Q3=1.708
Train size: 5594  |  Test size: 2398
```

```
# Tree 1: Binary Classification
clf_binary = tree.DecisionTreeClassifier(
    max_depth=4,
    min_samples_split=10,    # ← fixed (was 400)
    min_samples_leaf=5,      # ← fixed (was 200)
    random_state=RANDOM_STATE
)
clf_binary.fit(X_train, y_train_lbl)

y_pred_binary = clf_binary.predict(X_test)
print("=" * 55)
print("Tree 1 — Binary Classification")
print(f"  Test Accuracy : {accuracy_score(y_test_lbl, y_pred_binary):.4f}")
print(classification_report(y_test_lbl, y_pred_binary,
      target_names=["poor", "good"], zero_division=0))

# Tree 2: Regression
```

```
reg_tree = tree.DecisionTreeRegressor(
    max_depth=4,
    min_samples_split=10,    # ← fixed
    min_samples_leaf=5,      # ← fixed
    random_state=RANDOM_STATE
)
reg_tree.fit(X_train, y_train_scr)

y_pred_reg = reg_tree.predict(X_test)
print("=" * 55)
print("Tree 2 — Regression")
print(f"  Test MSE  : {mean_squared_error(y_test_scr, y_pred_reg):.4f}")
print(f"  Test RMSE : {np.sqrt(mean_squared_error(y_test_scr, y_pred_reg)):.4f}

# Tree 3: Multi-Class Classification
clf_multi = tree.DecisionTreeClassifier(
    max_depth=5,
    min_samples_split=10,    # ← fixed
    min_samples_leaf=5,      # ← fixed
    random_state=RANDOM_STATE
)
clf_multi.fit(X_train, y_train_multi)

y_pred_multi = clf_multi.predict(X_test)
print("=" * 55)
print("Tree 3 — Multi-Class Classification")
print(f"  Test Accuracy : {accuracy_score(y_test_multi, y_pred_multi):.4f}")
print(classification_report(y_test_multi, y_pred_multi,
      target_names=class_names_multi, zero_division=0))
print("=" * 55)
```

```
=======================================================
Tree 1 — Binary Classification
  Test Accuracy : 0.4921
              precision    recall  f1-score   support

        poor       0.00      0.00      0.00      1218
        good       0.49      1.00      0.66      1180

    accuracy                           0.49      2398
   macro avg       0.25      0.50      0.33      2398
weighted avg       0.24      0.49      0.32      2398

=======================================================
Tree 2 — Regression
  Test MSE  : 1.5577
  Test RMSE : 1.2481
=======================================================
Tree 3 — Multi-Class Classification
  Test Accuracy : 0.2364
              precision    recall  f1-score   support

       awful       0.00      0.00      0.00       645
```

```
        poor       0.00      0.00      0.00       573
        good       0.00      0.00      0.00       613
       great       0.24      1.00      0.38       567

    accuracy                           0.24      2398
   macro avg       0.06      0.25      0.10      2398
weighted avg       0.06      0.24      0.09      2398

==========================================================
```

```
# ── Tree 1: Binary Classification ──
plotTree(
    clf_binary,
    feature_names=feature_names,
    class_names=["poor", "good"],
    title="Tree 1: Binary Classification (good vs. poor)",
    save_path="tree_binary_classification.png",
    figsize=(28, 12)
)
```

```
Saved: tree_binary_classification.png
```

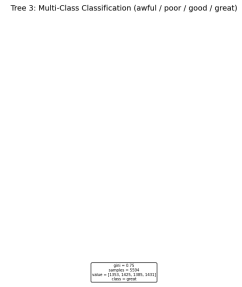
```
# ── Tree 2: Regression ──
plotTree(
    reg_tree,
    feature_names=feature_names,
    class_names=None,           # no class names for regression
    title="Tree 2: Regression (predicting Cello score)",
    save_path="tree_regression.png",
    figsize=(28, 12)
)
```

```
Saved: tree_regression.png
```

```
# ── Tree 3: Multi-Class ──
plotTree(
    clf_multi,
    feature_names=feature_names,
    class_names=class_names_multi,
    title="Tree 3: Multi-Class Classification (awful / poor / good / great)",
    save_path="tree_multiclass.png",
```

```
    figsize=(32, 14)
)
```

```
Saved: tree_multiclass.png
```




```
print("X shape:", X.shape)
print("Unique values in X:", np.unique(X))
print("Column variance (first 10 rules):")
print(pd.DataFrame(X, columns=feature_names).var().head(100))
print("\nLabel counts:", np.unique(y_labels, return_counts=True))
```

```
X shape: (7992, 5)
Unique values in X: [1]
Column variance (first 10 rules):
hw1_rules_1_Rule_N_v1     0.0
hw1_rules_Verification    0.0
hw1_rules_2_Rule_N_v1     0.0
hw1_rules_1_Rule_N        0.0
hw1_rules_2_Rule_N        0.0
dtype: float64
```

```
Label counts: (array([0, 1]), array([3996, 3996]))
```

## Extra Credit

```
true_df = pd.read_csv("true_scores.csv")
transformer_df = pd.read_csv("transformer_predicted_scores.csv")
print("True columns:", true_df.columns.tolist())
print("Transformer columns:", transformer_df.columns.tolist())
print(true_df.head(2))
print(transformer_df.head(2))
```

```
True columns: ['weight']
Transformer columns: ['weight']
     weight
0 -0.693147
1 -0.693147
     weight
0 -1.141758
1 -0.182603
```

```
# True scores
purity_true, designToRule_true = ruleEvaluateByGroup(
    "eval_true", "hw1_true", "hw1_rules", "median"
)
print("True eval done:", purity_true.shape)
```

```
True eval done: (5, 14)
```

```
# Transformer scores
purity_transformer, designToRule_transformer = ruleEvaluateByGroup(
    "eval_transformer", "hw1_transformer", "hw1_rules", "median"
)
print("Transformer eval done:", purity_transformer.shape)
```

```
Transformer eval done: (5, 14)
```

```
# Merge all three
purity_comparison = purity_true[["ruleSpaceID", "impact"]].rename(
    columns={"impact": "true_impact"}
).merge(
    purity_metrics_df[["ruleSpaceID", "impact"]].rename(
        columns={"impact": "gnn_impact"}),
    on="ruleSpaceID"
).merge(
    purity_transformer[["ruleSpaceID", "impact"]].rename(
        columns={"impact": "transformer_impact"}),
    on="ruleSpaceID"
```

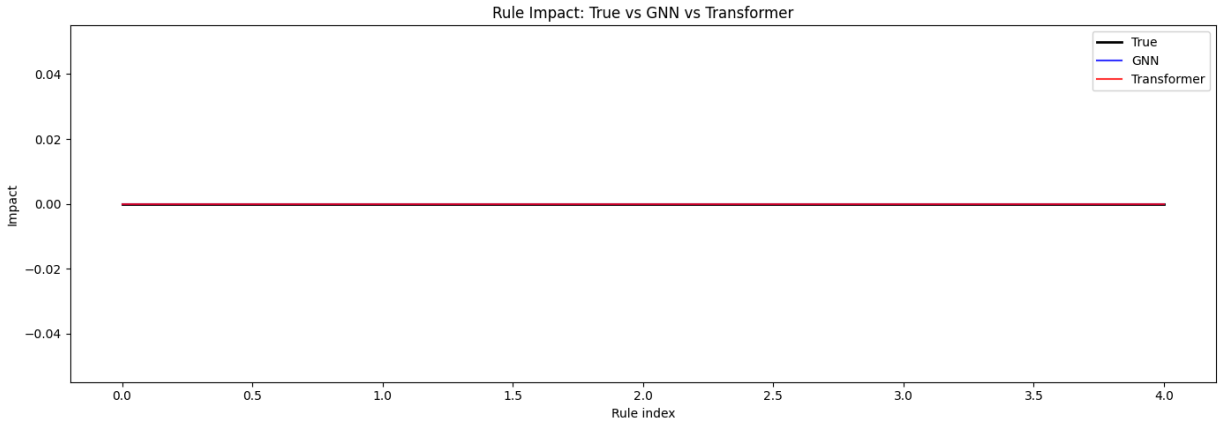
```
)

# Quantify
diff_gnn   = np.sum(np.abs(
    purity_comparison["gnn_impact"] - purity_comparison["true_impact"]))
diff_trans = np.sum(np.abs(
    purity_comparison["transformer_impact"] - purity_comparison["true_impact"]))


winner = 'GNN' if diff_gnn < diff_trans else 'Transformer'
print(f"\n→ {winner} more closely emulates true grammar")
```

```
→ Transformer more closely emulates true grammar
```

```
# Plot
fig, ax = plt.subplots(figsize=(14, 5))
x = range(len(purity_comparison))
ax.plot(x, purity_comparison["true_impact"],
        label="True", linewidth=2, color="black")
ax.plot(x, purity_comparison["gnn_impact"],
        label="GNN", linewidth=1.5, color="blue", alpha=0.8)
ax.plot(x, purity_comparison["transformer_impact"],
        label="Transformer", linewidth=1.5, color="red", alpha=0.8)
ax.set_xlabel("Rule index")
ax.set_ylabel("Impact")
ax.set_title("Rule Impact: True vs GNN vs Transformer")
ax.legend()
plt.tight_layout()
plt.savefig("purity_comparison.png", dpi=150)
plt.show()
```


Rule Impact: True vs GNN vs Transformer
True
GNN
Transformer
Impact
0.04
0.02
0.00
−0.02
−0.04
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Rule index